

WorkshopPlus: Azure AI Platform and Services

Introduction to LLM Fine-tuning

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Learning Units covered in this Module



Fine-Tuning Overview



No-Code Fine-tuning



Fine-Tuning Overview

Objectives

After completing this Learning, you will be able to:

1

Understand the fundamentals of fine-tuning

- ✓ What's Model Fine-tuning?
- ✓ Why is Fine-tuning used?
- ✓ When to apply Fine-tuning?
- ✓ Fine-Tuning Methodologies
- ✓ Fine-Tuning Evaluations & Process Monitoring



What, Why and When

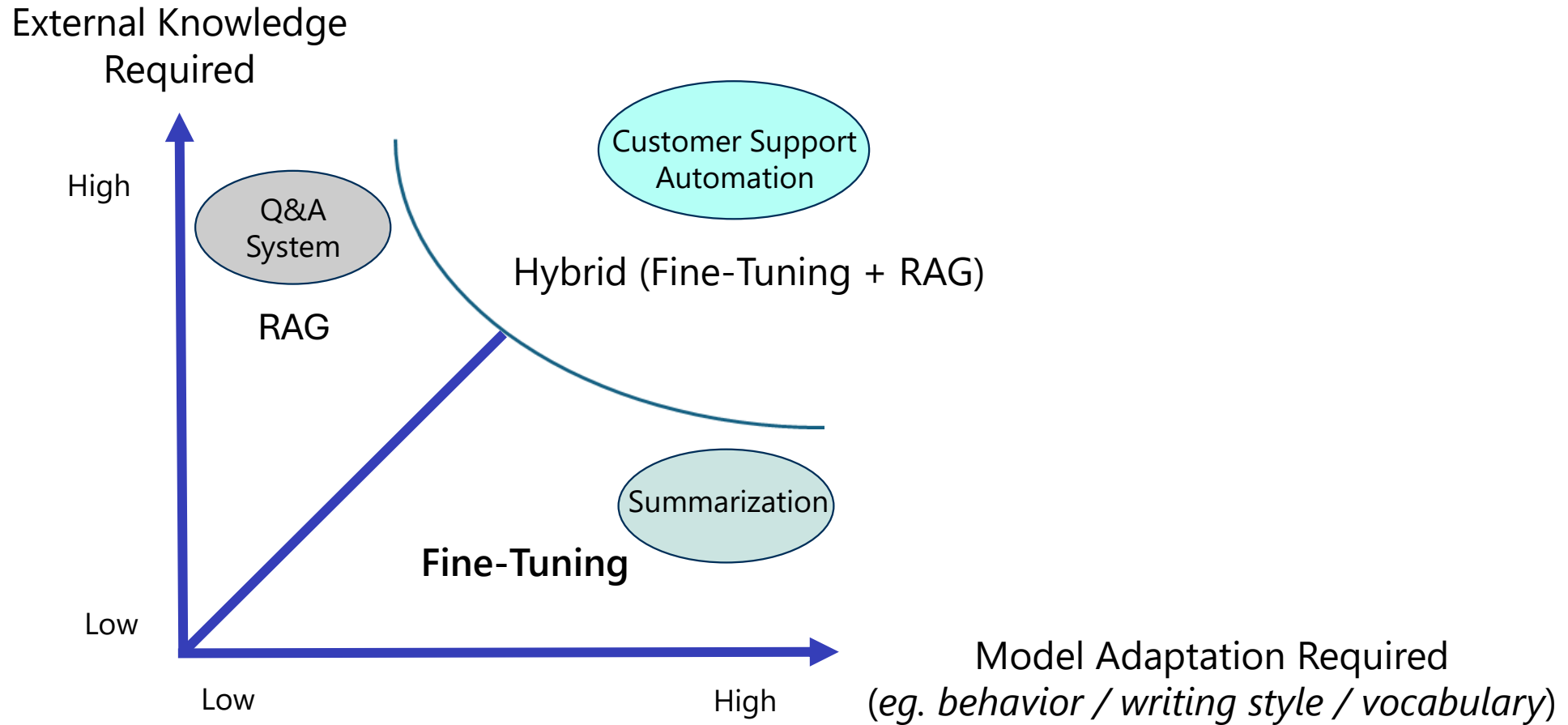
What *is* Fine-Tuning?

Fine-Tuning refers to **customizing a pre-trained LLM** with additional training on a specific task or new dataset for enhanced performance, new skills, or improved accuracy.



A popular fine-tuning method used by Azure OpenAI is called [Low-Rank Adaptation](#) (LoRA). LoRA works by approximating the original high-rank matrix with a lower rank one, only fine-tuning a smaller subset of "important" parameters. This technique reduces the complexity of fine tuning while maintaining performance, making training faster and more affordable.

Where does Fine-Tuning fit in?



Why to customize GenAI models?

Scalability

Customization enables AI models to scale and adapt to specific enterprise needs.

Reducing hallucinations

Tailored models are less likely to produce inaccurate or irrelevant responses.

Increased reliability

Enhances the model's accuracy for domain-specific tasks.

Improved efficiency

Customization ensures faster and more precise results, saving time and resources.

Tailored solutions

Models are fine-tuned for specific use cases, providing more relevant and context-aware outcomes.

General Purpose Use Cases



Reducing prompt length



Teaching new skills



Improving tool use



Domain adaptation

Vertical Specific Use Cases



Natural language to code



Translation and dialects



Style and formatting



Customer specific knowledge infusion

Fine-Tuning Considerations

Advantages

- Improves results for specific domains, styles, formats where generic models fall short.
- Trains on more examples than a prompt allows, improving robustness and task accuracy.
- Reduces token usage and prompt length for lower costs and faster responses.
- Combines well with strong initial prompt engineering efforts.

Limitations

- Depends on a well-prepared, high-quality labelled dataset.
- Longer iteration cycles due to data preparation and training.
- Solution is tied to a specific base model version. Might become obsolete after an improved base model is released.
- The fine-tuned models may overfit to the training data, making them less generalizable to unseen tasks.

Azure Fine-Tuning Options



Azure OpenAI

Fully managed service - Only option for AOAI fine-tuning

GPT-3.5-turbo,
GPT-4/4o/4o-mini

East US2, NC US,
Sweden Central

- Fine-tuning billed per token.
- Model host billed per hour.
- Inference is billed per token.



Foundry Serverless FT (MaaS)

Fully managed fine-tuning - no need for own GPU capacities

Llama3, Phi3, Mistral, NTT

East US2, West US3

- Fine-tuning jobs and model hosting is billed by the hour.
- Inference is billed per token.



Foundry Managed FT (MaaS)

Bring your own GPU capacity for fine-tuning and inference

Supported for a larger number of models

All regions supported

- Fine-tuning jobs and inference are billed for GPUs.



AzureML

Bring your own GPU capacity for fine-tuning and inference

All models including custom architectures

All regions supported

- Fine-tuning jobs and inference are billed for GPUs.

Complexity & More Knowledge Required →

Note that this overview does not include Azure AI service specific fine-tuning options (e.g., the Custom Neural Voice training).

Fine-Tuning Methods

Azure Fine-Tuning Methods

Supervised Fine-Tuning

- GPT-35-Turbo (0125) – 16K context
- GPT-4o (0806) – 128K context
- GPT-4o-mini (0718) – 64k context
- GPT-4.1
- GPT-4.1-mini
- GPT-4.1-nano

Preference Fine-Tuning (DPO)

- GPT-4o (0806)

Reinforcement Fine-Tuning

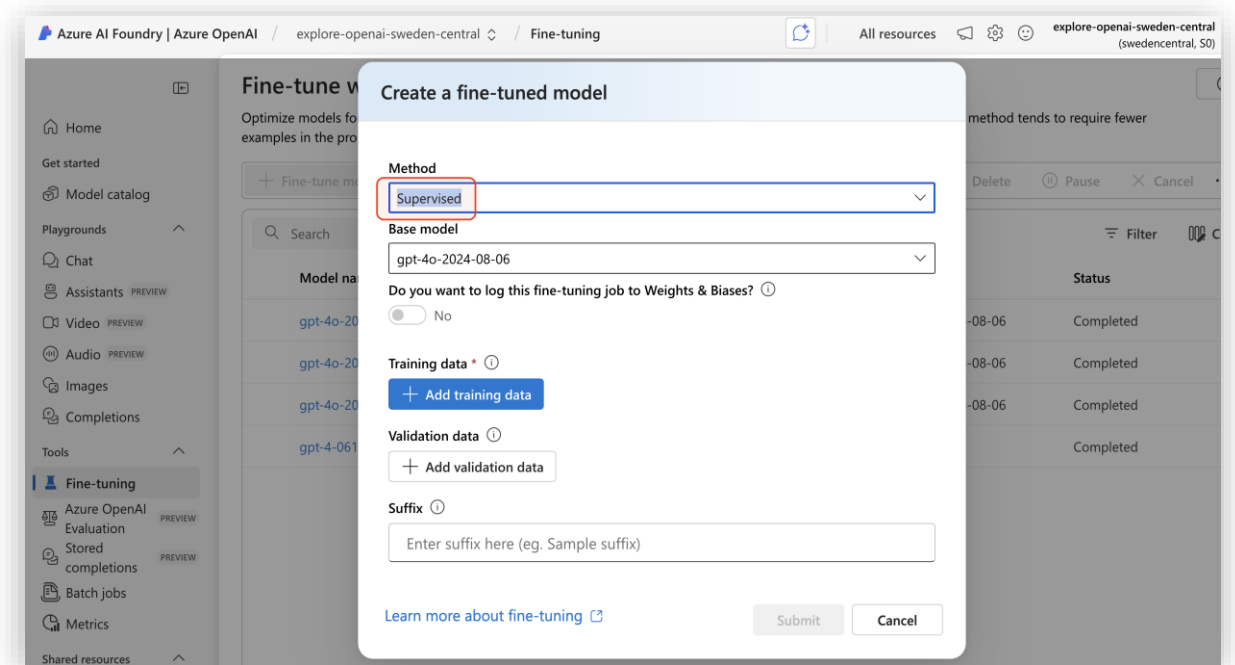
- o4-mini

Regional

Global

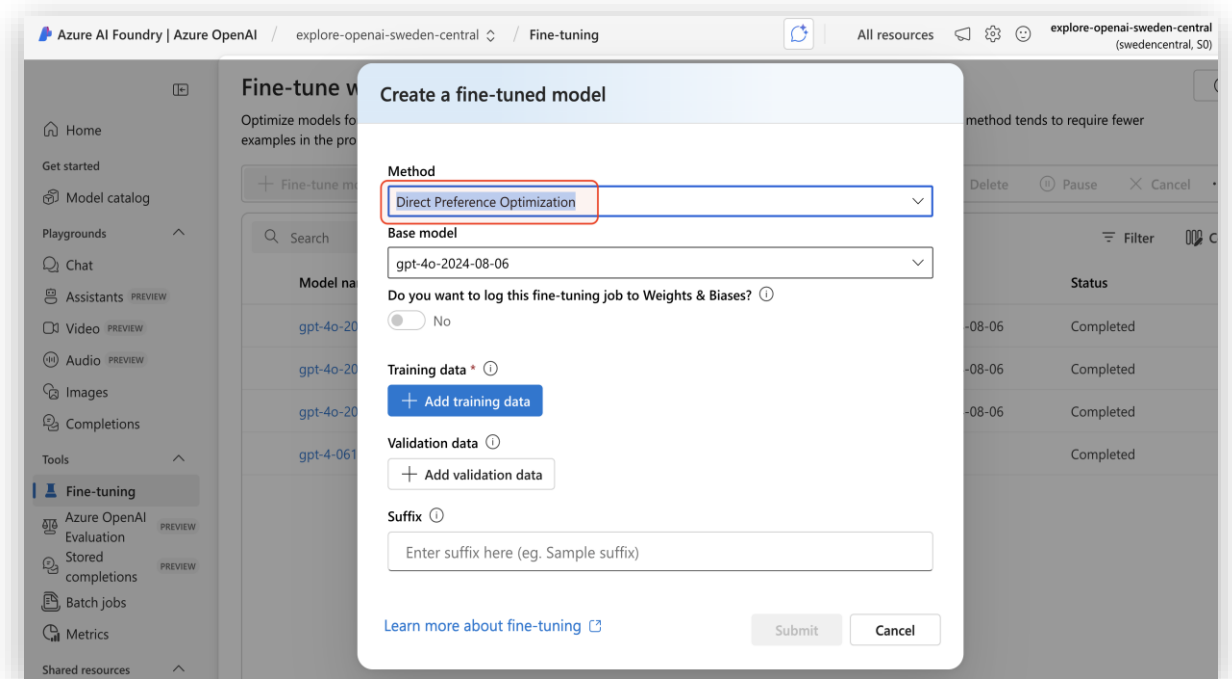
Supervised Fine-Tuning

- Aligns a model's responses with specific user prompts and preferred outputs.
- Most widely adopted method for tailoring model behaviour.
- Supported by most Azure OpenAI GPT models, including the GPT-4o and GPT-4.1 series.



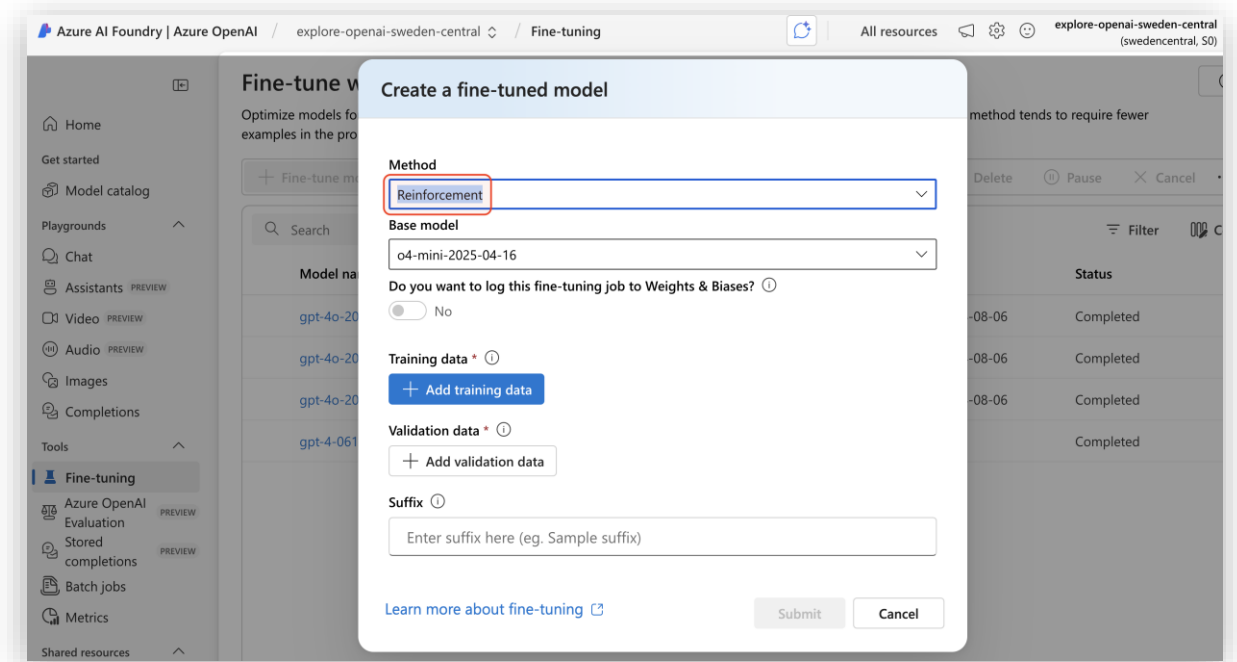
Preference Fine-Tuning (Direct Preference Optimization)

- Aligns a model's responses using user-labeled examples of preferred vs. non-preferred outputs.
- Can be combined with Supervised Fine-Tuning for a better personalization.
- Currently supported only on the GPT-4o model in Azure OpenAI.



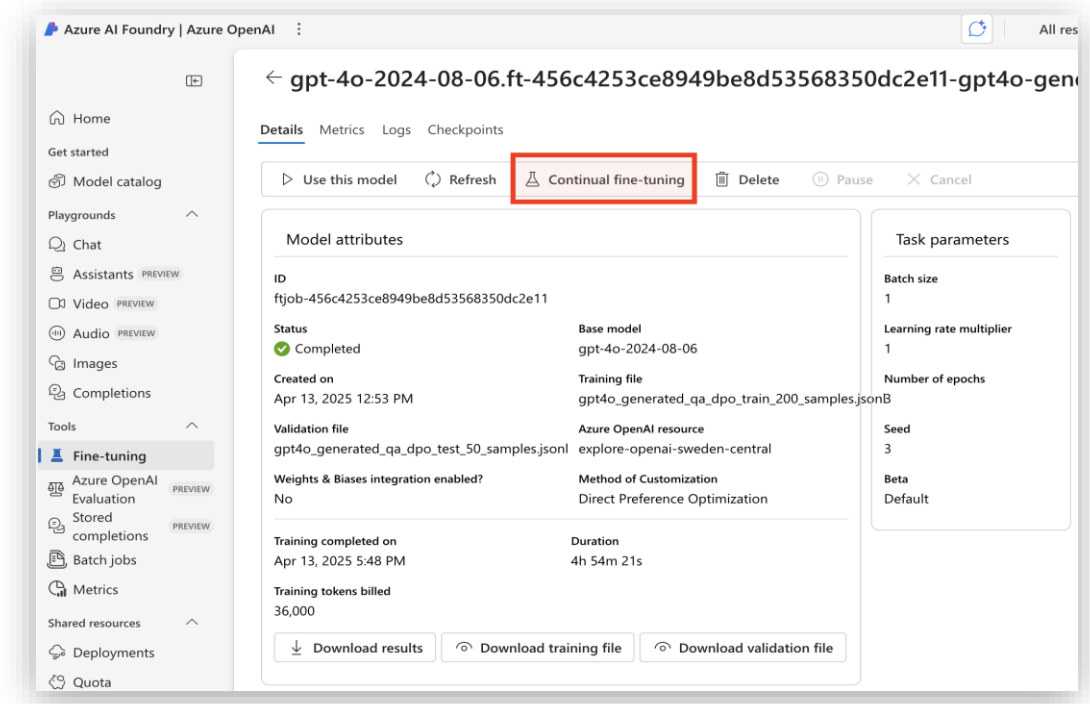
Reinforcement Fine-Tuning

- Aligns a model's responses via reward signals that encourage accurate reasoning and domain-specific behavior.
- Uses a grader or reward model to reinforce correct logic and penalize incorrect outputs.
- Currently only supported on the o4-mini model in Azure OpenAI.



Continual Fine-Tuning

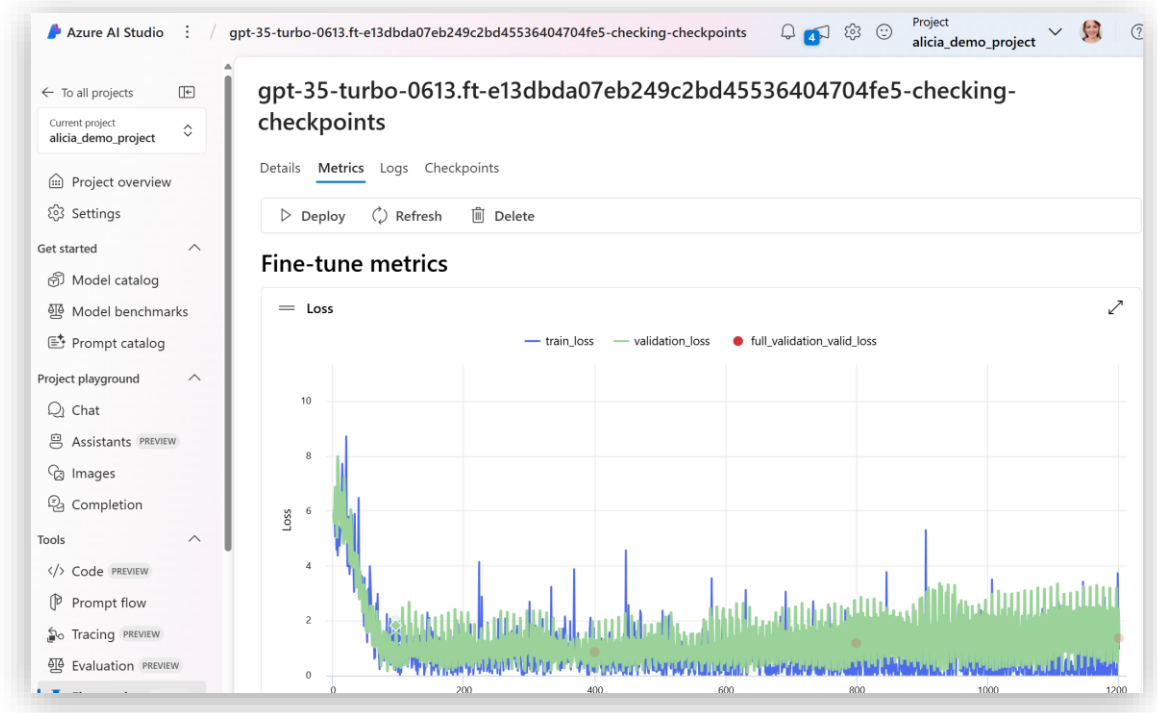
- **Update a fine-tuned model with new data** or continue from where you left off.
- Keep most parameters the same but **consider reducing the learning rate.**



Fine-Tuning Evaluations

Full Validation Statistics

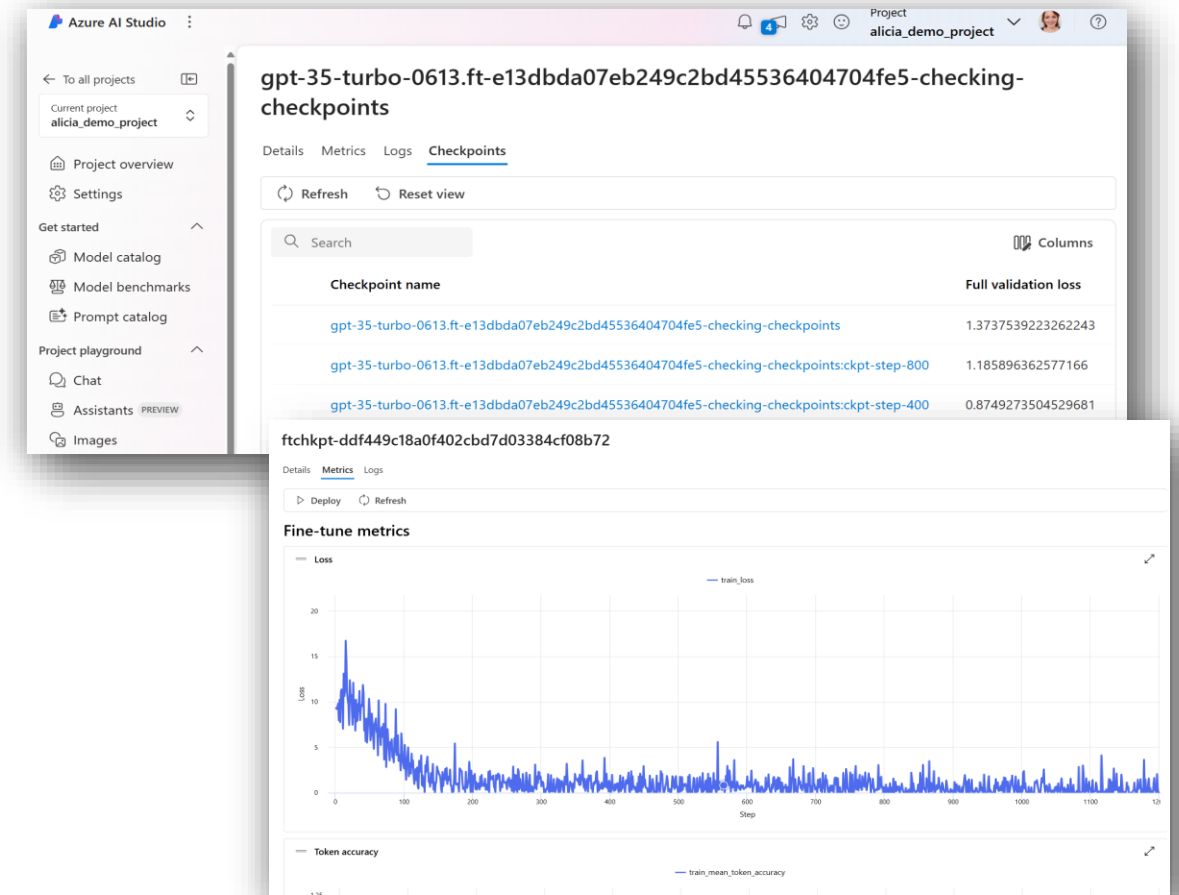
- **Validation data** helps measure how well your fine-tuning job performed and check if you've overfit.
- **Most** validation results are sampled from the full validation file.
- **Full Validation Statistics** are produced at the end of each epoch to give you a clearer picture.



Model Checkpoints

(it's ok if you accidentally overfit)

- A “**checkpoint**” is a **deployable model** created at the end of a training epoch.
- Using the UI or API you can **select, review, and deploy checkpoints** just like any other fine-tuned model.
- **The last three checkpoints** are automatically saved.



Deploy Fine-Tuned Model – Multiple Types

Pay-as-you-Go (PayGo)

- **Standard** - the existing regional deployment model
- **Global Standard** - deploy from any training region at **reduced per-token rates** in exchange for relaxed data residency.

Provisioned Managed (PTU-M)

- Predictable performance with reserved capacity, currently offered regionally.

The screenshot shows the 'Deploy model' configuration window for the model 'gpt-4o-mini-2024-07-18.ft-cec320a659f148849ddd10fe2e529d8d-test'. The 'Deployment name' is 'gpt-4o-mini-2024-07-18-test'. The 'Deployment type' dropdown is set to 'Provisioned-managed', with a list of options including 'Standard', 'Global Standard (Preview)', and 'Provisioned-managed'. A status message indicates '300 PTUs of quota are available for this region and model.' Below this, a slider for 'Provisioned throughput units (PTUs)' is shown at 25/300. The 'Content filter' is set to 'DefaultV2'. At the bottom right are 'Confirm pricing' and 'Cancel' buttons.

Deploy model gpt-4o-mini-2024-07-18.ft-cec320a659f148849ddd10fe2e529d8d-test

Deployment name *

gpt-4o-mini-2024-07-18-test

Deployment type

Provisioned-managed

Standard

Global Standard (Preview)

Provisioned-managed

300 PTUs of quota are available for this region and model.

Provisioned throughput units (PTUs)

25/300

Content filter

DefaultV2

Confirm pricing Cancel

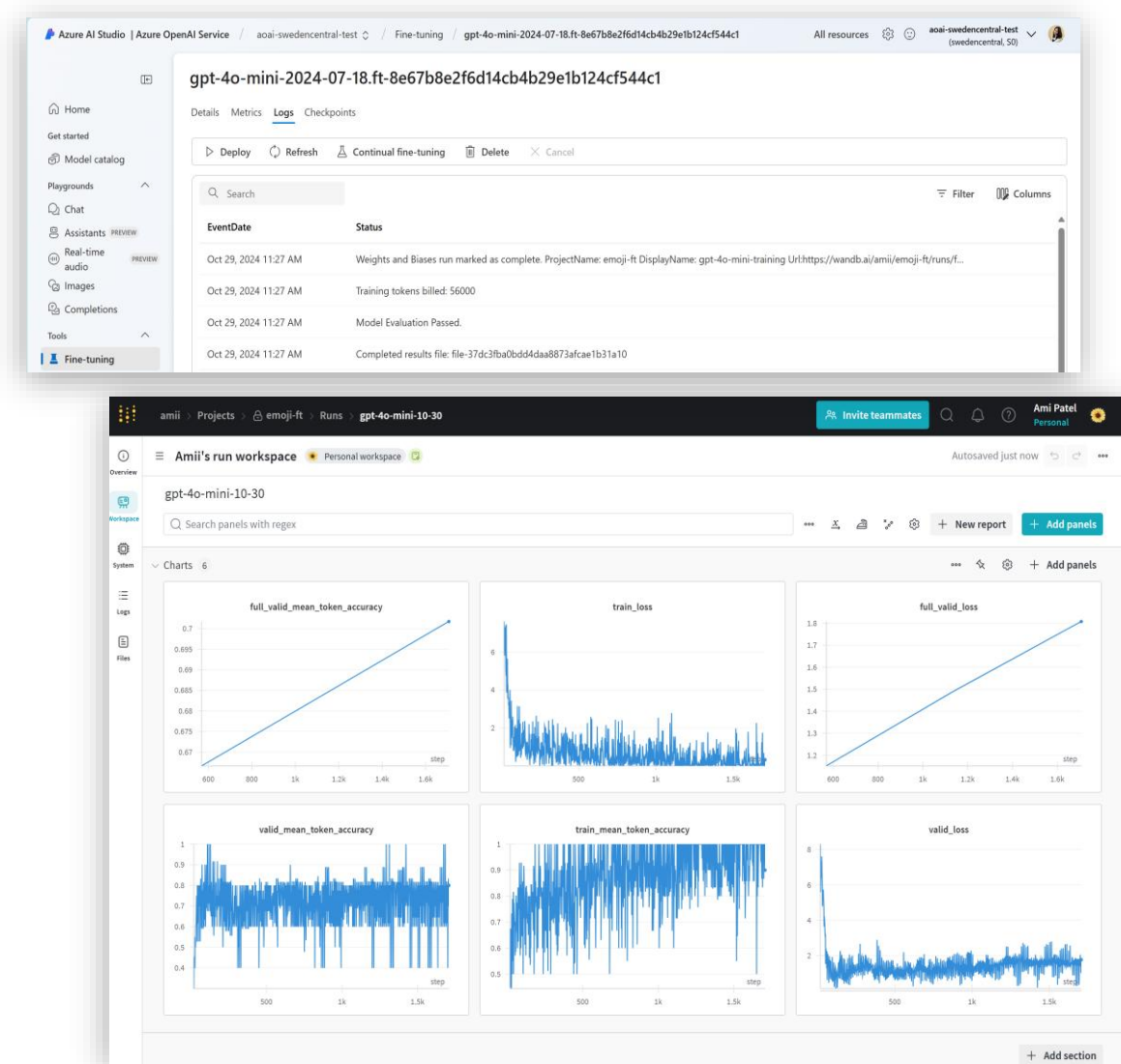
Weights & Biases Integration

What is this?

- Use **Weights & Biases** powerful capabilities for **experiment tracking, visualization, model management, and collaboration**.
- With this integration, users can utilize **W&B Weave** to evaluate, monitor, and iterate on the performance of their fine-tuned models from **Azure OpenAI service** in real-time.

How to use?

- Set up an Azure Key Vault to store your W&B API key.
- Enable the integration in Azure OpenAI Studio using **Manage Integrations** in fine-tuning.
- After successfully adding this integration, proceed to fine-tuning a model and toggle on "Do you want to log this fine-tuning job to Weights & Biases?"
- Use the link in Studio to navigate to your W&B project.



Packaging & Pricing

Azure OpenAI Charges for:

- Training - per 1K tokens (same as OpenAI).
- Hosting - depends on the model, most are \$1.70/hour.
- Inferencing - same as base model

AOAI is cheaper for high TPM workloads, compared to OpenAI.

- Break even around 7K TPM: avg TPM is 11K, thus AOAI is more favorable.
- *Save Big on Fine Tuning with Azure OpenAI (microsoft.com)*

We offer regional training and inferencing.

- Easy for customers to comply with regulations like HIPAA, GDPR, etc.

Production scale workloads are supported with provisioned capacity.

- Same price as base models: customers pay for PTUs - not hosting and inferencing.

No Code Fine-Tuning

Objectives

After completing this Learning, you will be able to:

1

Understand the tools and methods

- ✓ AI Foundry Dashboard: a no-code platform for managing fine-tuning tasks.

2

Understand fine-tuning process

- ✓ AOAI Model Fine-tuning using Azure AI Foundry
- ✓ Open-Source Model Fine-tuning using Model Catalog and Azure AI Foundry



Fine-Tuning via Azure AI Foundry Dashboard

✓ Prerequisites

- Azure subscription & AI Foundry project
- Azure OpenAI connection (*in supported region*)
- Supported models:
gpt-35-turbo-0125, gpt-4o-0806, gpt-4o-mini, gpt-4.1-mini, gpt-4.1-nano, o4-mini
- Training data: ≥50 samples in .jsonl UTF-8 format
- (*Optional*) Validation data

🔧 Steps Overview

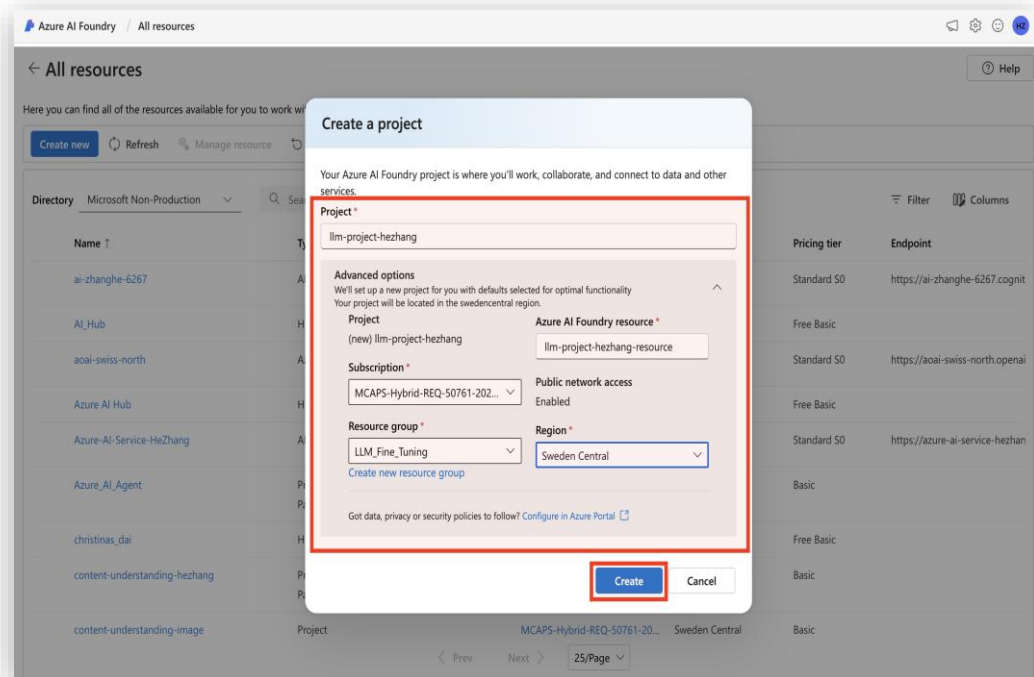
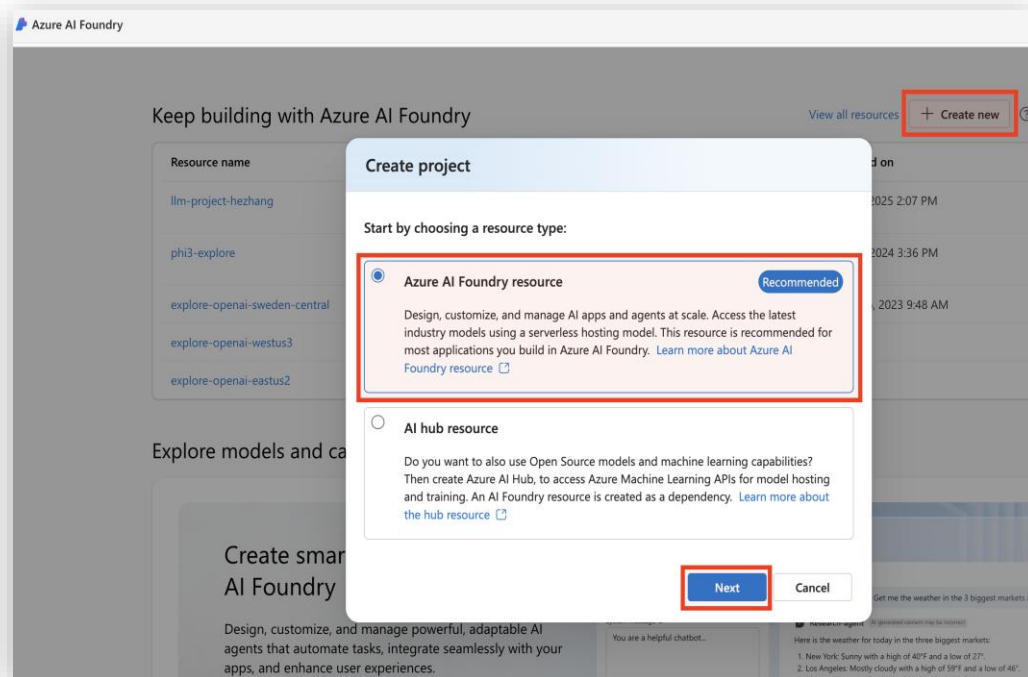
1. Create a Project in Azure AI Foundry
2. Launch Fine-tune Wizard
3. Select Base Model (*e.g., gpt-4.1-mini*)
4. Upload Training Data (*Supervised, DPO, or RL*)
5. (*Optional*) Add Validation Data
6. (*Optional*) Configure Advanced Options
7. Review & Submit
8. Deploy the fine-tuned model
9. Test & Use via Playground or API
10. (*Optional*) Clean up resources
11. (*Optional*) Continuous fine-tuning

<https://learn.microsoft.com/en-us/azure/ai-services/openai/how-to/fine-tuning?tabs=azure-openai&pivots=programming-language-studio>

Fine-Tuning via Azure AI Foundry Dashboard

Fine-Tuning Azure OpenAI GPT Models

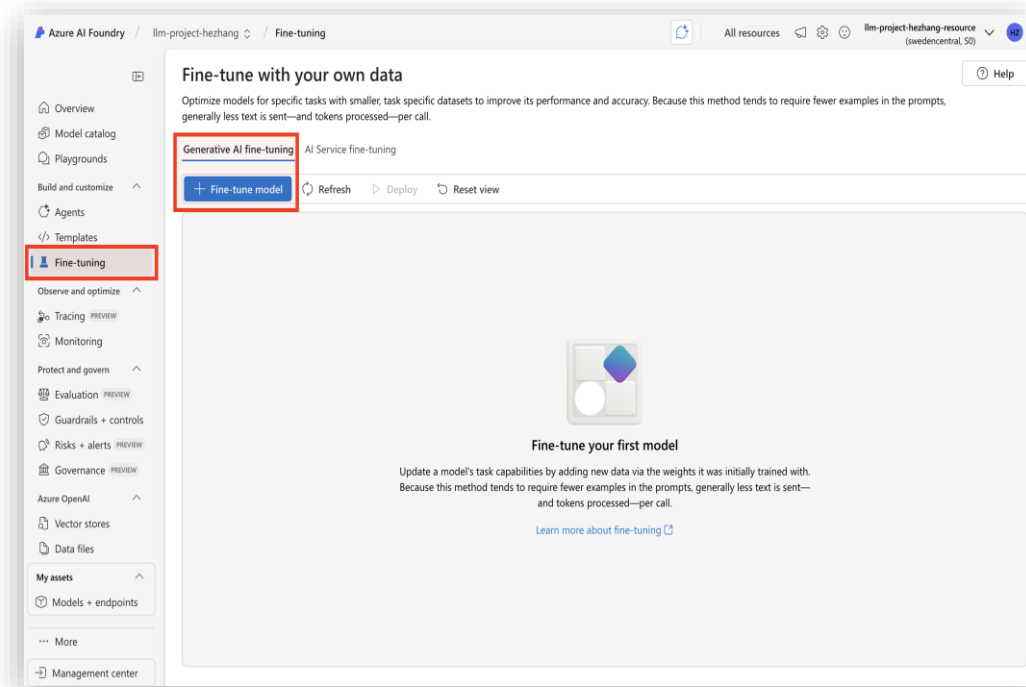
Step 1: Create a Project in Azure AI Foundry



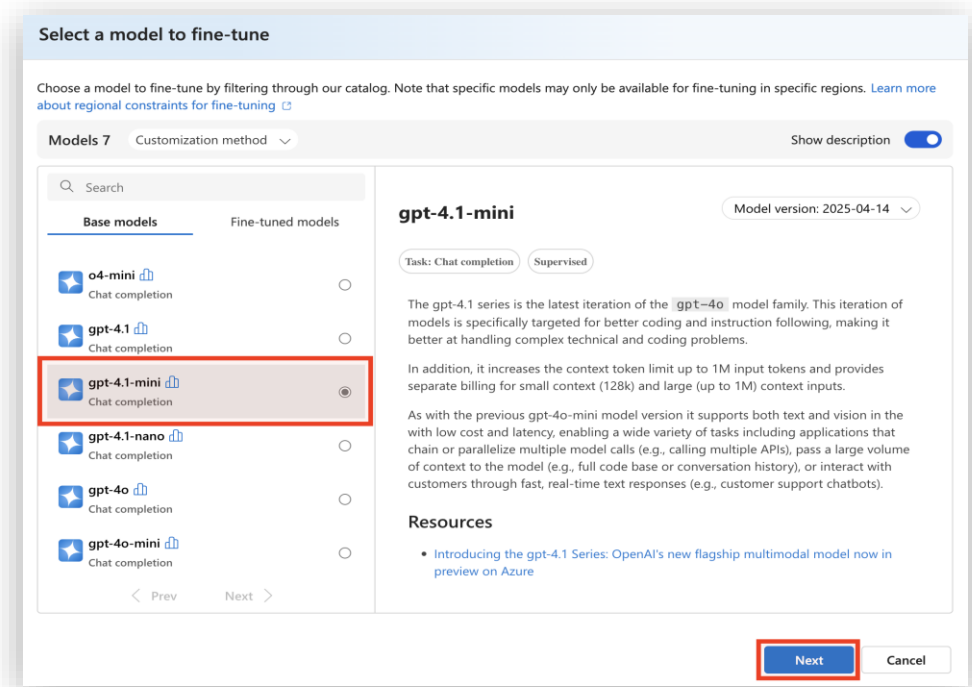
Fine-Tuning via Azure AI Foundry Dashboard

Fine-Tuning Azure OpenAI GPT Models

Step 2: Launch *Fine-tune* Wizard



Step 3: Select *Base model*



Fine-Tuning via Azure AI Foundry Dashboard

Fine-Tuning Azure OpenAI GPT Models

Step 4: Upload Training Data

Create a fine-tuned model

Method

Supervised

Base model

gpt-4.1-mini-2025-04-14

Training type

Standard

Standard training offers [data residency](#) guarantees. No data leaves the current resource region.

Training data *

+ Add training data

Validation data

+ Add validation data

Suffix

Enter suffix here (eg. Sample suffix)

Seed

Random

[Learn more about fine-tuning](#)

Submit Cancel

Fine-tune gpt-4.1-mini

Choose a dataset to customize your model. Ensure your training data is in a .jsonl file, formatted for chat completions, with 'messages' as the row header.

Get started faster - learn how to create datasets using these [samples](#) on our [GitHub repo](#)

Training data *

Upload files

Choose a file to upload from your local drive. File must be under 200 MB.

Upload file

☐ Overwrite if already exists

Upload list

training_set_10samples.jsonl	2.64 KB	...
------------------------------	---------	-----

☐ Skip preview

Preview of top 3 rows from your dataset (Total: 10 rows)

messages

```
[{"role": "system", "content": "Clippy is a factual chatbot that is also sarcastic."}, {"role": "user", "content": "What is the capital of France?"}, {"role": "assistant", "content": "The capital of France is Paris."}]
```

[Learn more about preparing your own data](#)

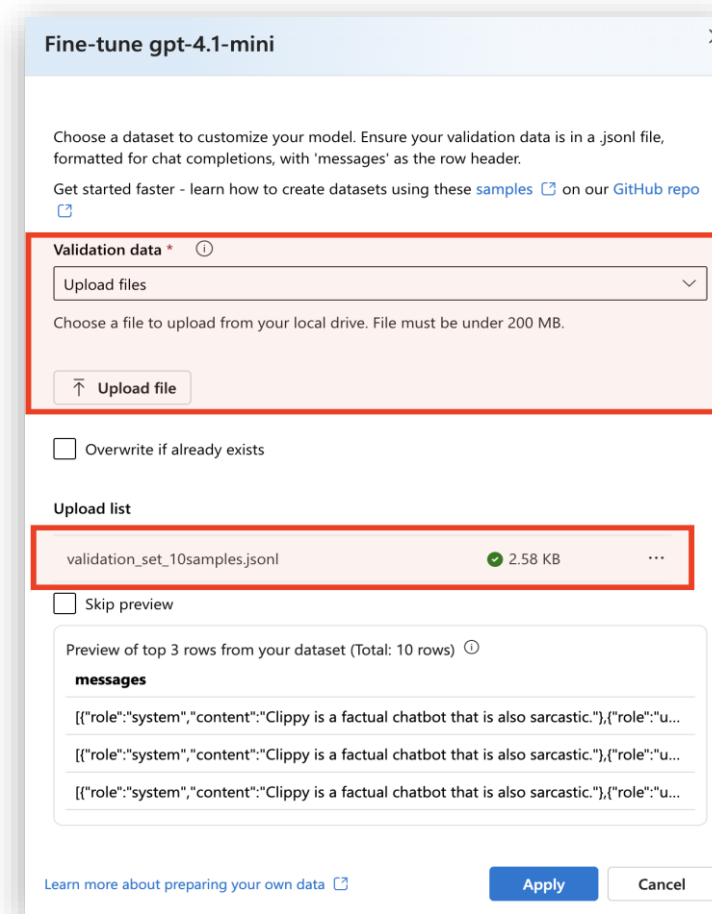
Apply Cancel

* One can refer to the MS Learn Document [here](#) for required fine-tuning file format.

Fine-Tuning via Azure AI Foundry Dashboard

Fine-Tuning Azure OpenAI GPT Models

Step 5: Add Validation Data (Optional)



Fine-tune gpt-4.1-mini

Choose a dataset to customize your model. Ensure your validation data is in a .jsonl file, formatted for chat completions, with 'messages' as the row header.

Get started faster - learn how to create datasets using these [samples](#) on our [GitHub repo](#)

Validation data *

Upload files

Choose a file to upload from your local drive. File must be under 200 MB.

Upload file

☐ Overwrite if already exists

Upload list

validation_set_10samples.jsonl	2.58 KB	...
--------------------------------	---------	-----

☐ Skip preview

Preview of top 3 rows from your dataset (Total: 10 rows)

messages

```
[{"role": "system", "content": "Clippy is a factual chatbot that is also sarcastic."}, {"role": "u..."}]
```

```
[{"role": "system", "content": "Clippy is a factual chatbot that is also sarcastic."}, {"role": "u..."}]
```

```
[{"role": "system", "content": "Clippy is a factual chatbot that is also sarcastic."}, {"role": "u..."}]
```

[Learn more about preparing your own data](#)

Apply Cancel

* One can refer to the MS Learn Document [here](#) for required fine-tuning file format.

Fine-Tuning via Azure AI Foundry Dashboard

Fine-Tuning Azure OpenAI GPT Models

Step 6: Configure Advanced Options (Optional)

Create a fine-tuned model

Training data ⓘ
training_set_10samples_jsonl_2025-06-08_062829.UTC
Edit 🗑️

Validation data ⓘ
validation_set_10samples_jsonl_2025-06-08_062904.UTC
Edit 🗑️

Suffix ⓘ
gpt-4-1-mini-ft-10samples-qa

Seed ⓘ
1

Configure hyperparameters ⓘ

☐ Batch size ⓘ
☐ Learning rate multiplier ⓘ
☐ Number of epochs ⓘ

Default (Auto)
Default (Auto)
Default (Auto)

Automatically deploy when fine-tuning is complete (Preview)
☒ No

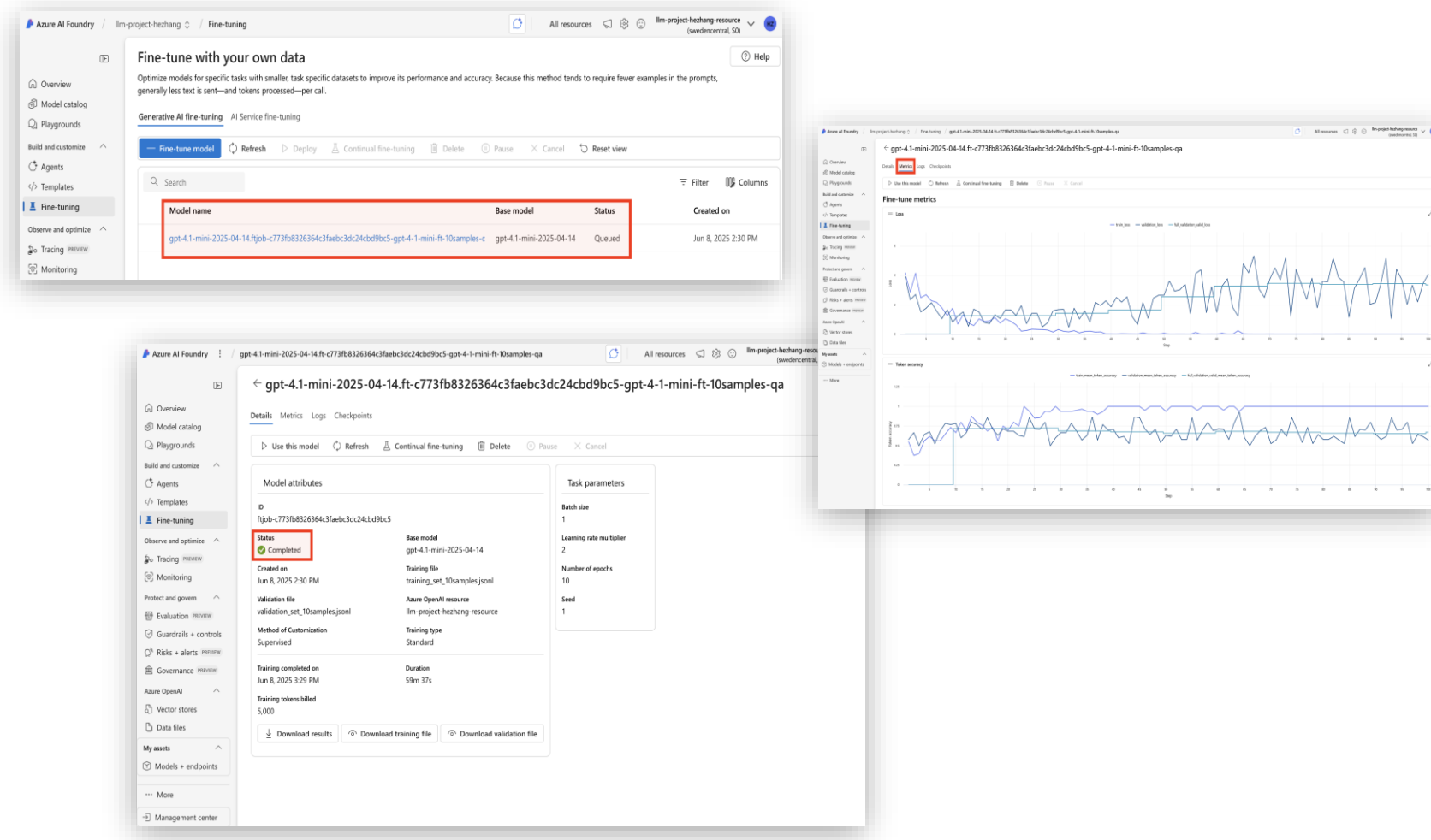
[Learn more about fine-tuning](#) Submit Cancel

Name	Type	Description
batch_size	integer	The batch size to use for training. The batch size is the number of training examples used to train a single forward and backward pass. In general, we've found that larger batch sizes tend to work better for larger datasets. The default value as well as the maximum value for this property are specific to a base model. A larger batch size means that model parameters are updated less frequently, but with lower variance. When set to -1, batch_size is calculated as 0.2% of examples in training set and the max is 256.
learning_rate_multiplier	number	The learning rate multiplier to use for training. The fine-tuning learning rate is the original learning rate used for pre-training multiplied by this value. Larger learning rates tend to perform better with larger batch sizes. We recommend experimenting with values in the range 0.02 to 0.2 to see what produces the best results. A smaller learning rate may be useful to avoid overfitting.
n_epochs	integer	The number of epochs to train the model for. An epoch refers to one full cycle through the training dataset. If set to -1, the number of epochs is determined dynamically based on the input data.
seed	integer	The seed controls the reproducibility of the job. Passing in the same seed and job parameters should produce the same results, but may differ in rare cases. If a seed isn't specified, one will be generated for you.
Beta	integer	Temperature parameter for the dpo loss, typically in the range 0.1 to 0.5. This controls how much attention we pay to the reference model. The smaller the beta, the more we allow the model to drift away from the reference model. As beta gets smaller the more, we ignore the reference model.

Fine-Tuning via Azure AI Foundry Dashboard

Fine-Tuning Azure OpenAI GPT Models

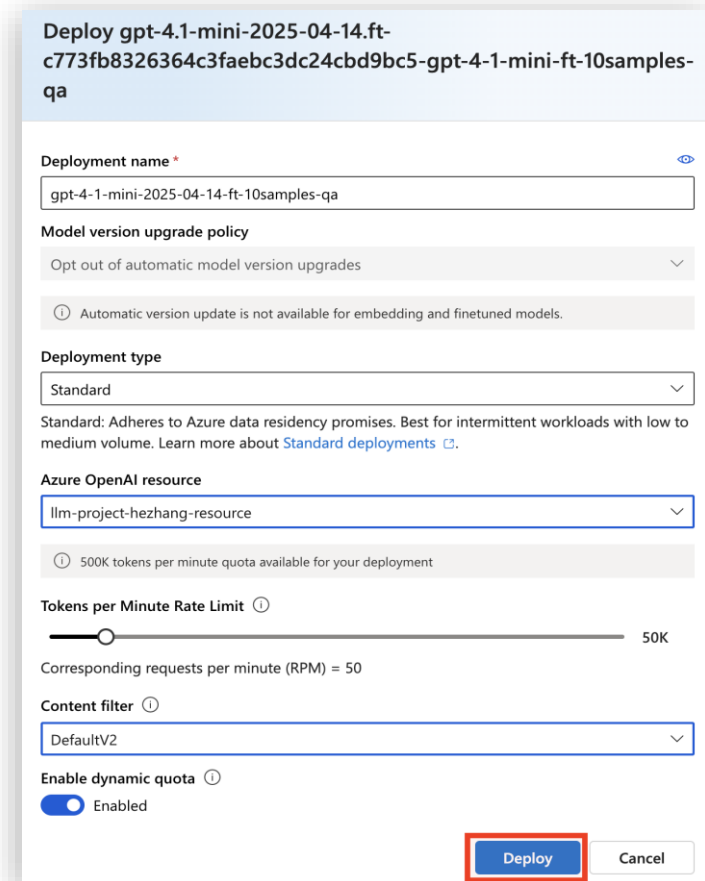
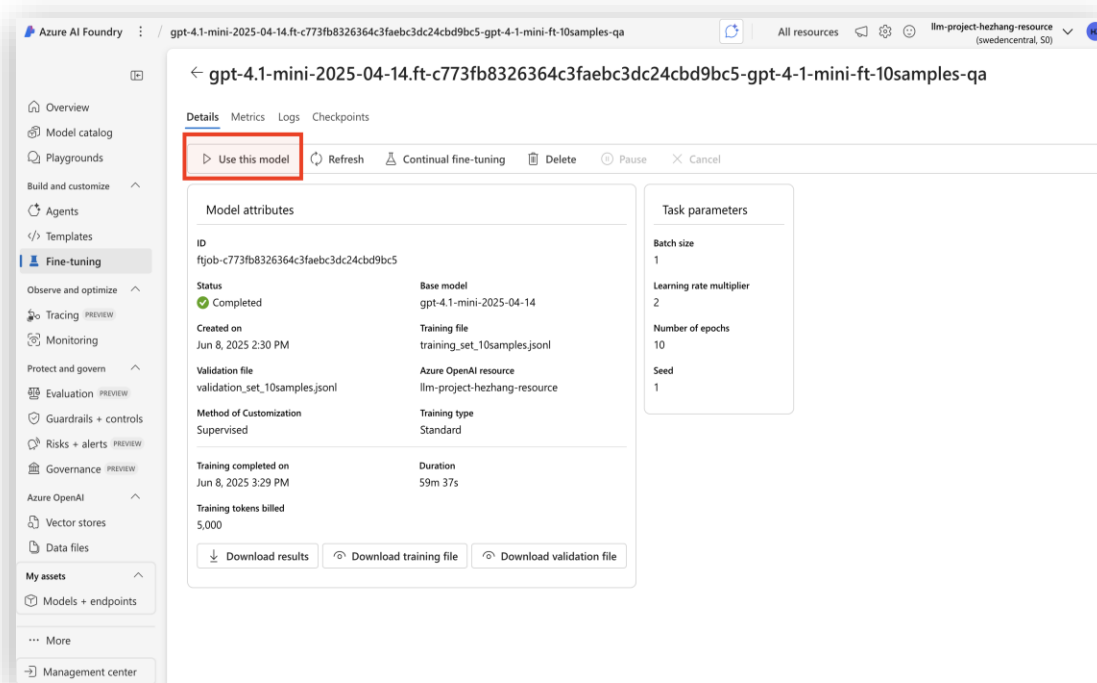
Step 7: Review & Submit



Fine-Tuning via Azure AI Foundry Dashboard

Fine-Tuning Azure OpenAI GPT Models

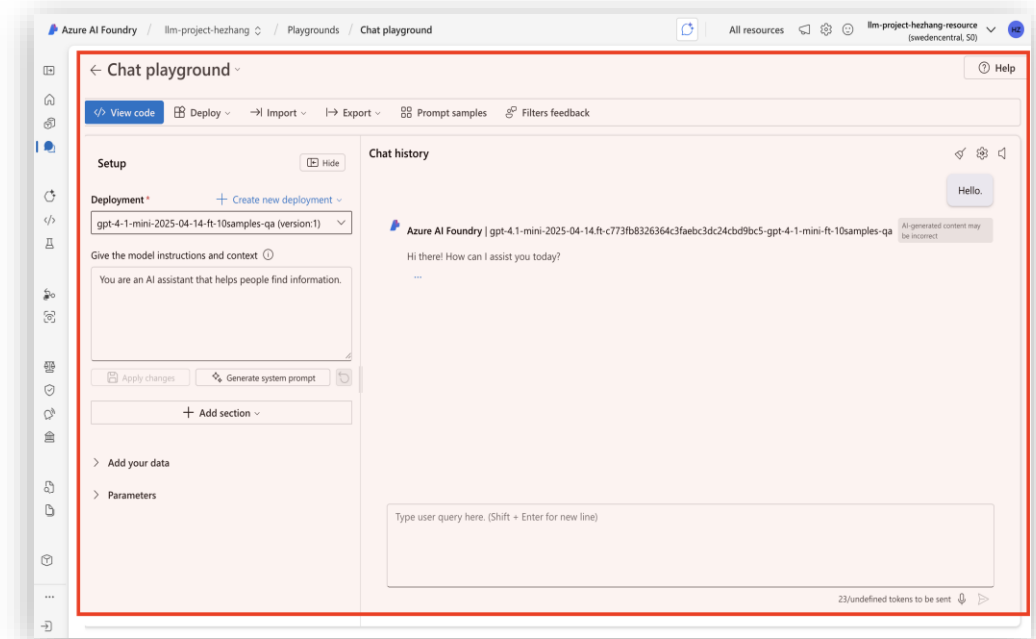
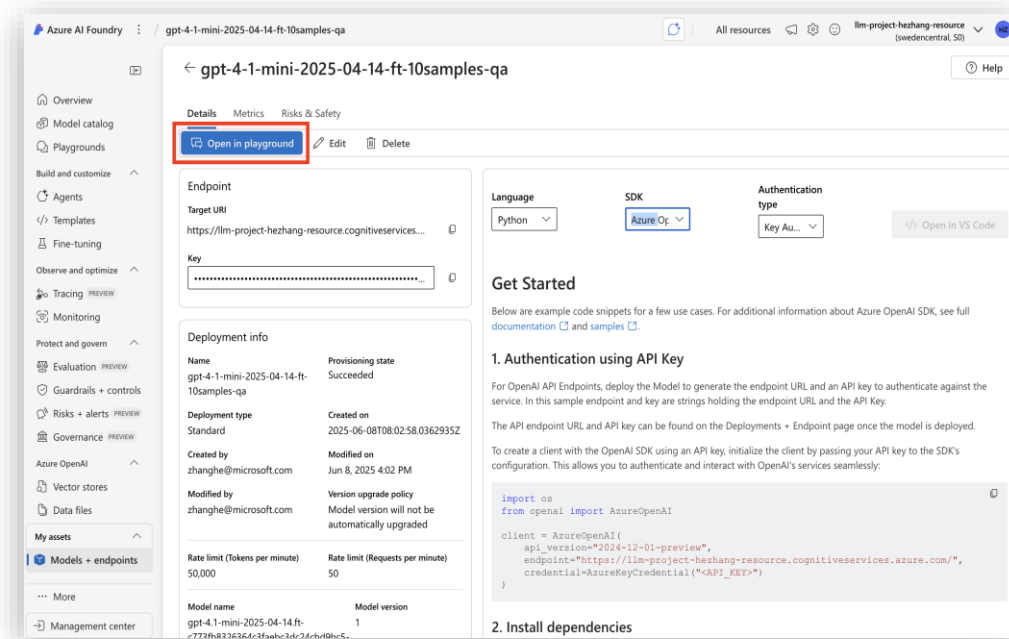
Step 8: Deploy The Fine-Tuned Model



Fine-Tuning via Azure AI Foundry Dashboard

Fine-Tuning Azure OpenAI GPT Models

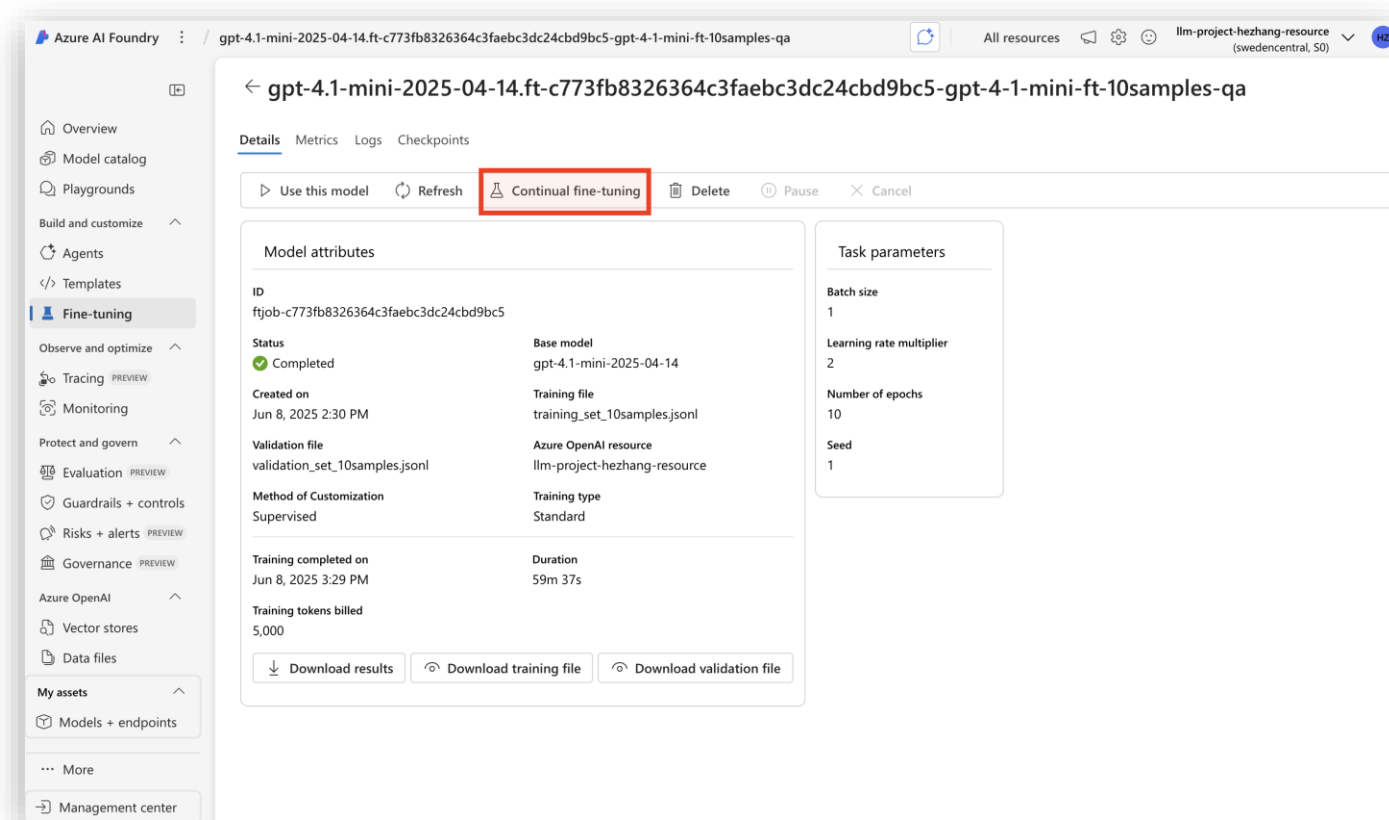
Step 9: Test & Use via Playground or API



Fine-Tuning via Azure AI Foundry Dashboard

Fine-Tuning Azure OpenAI GPT Models

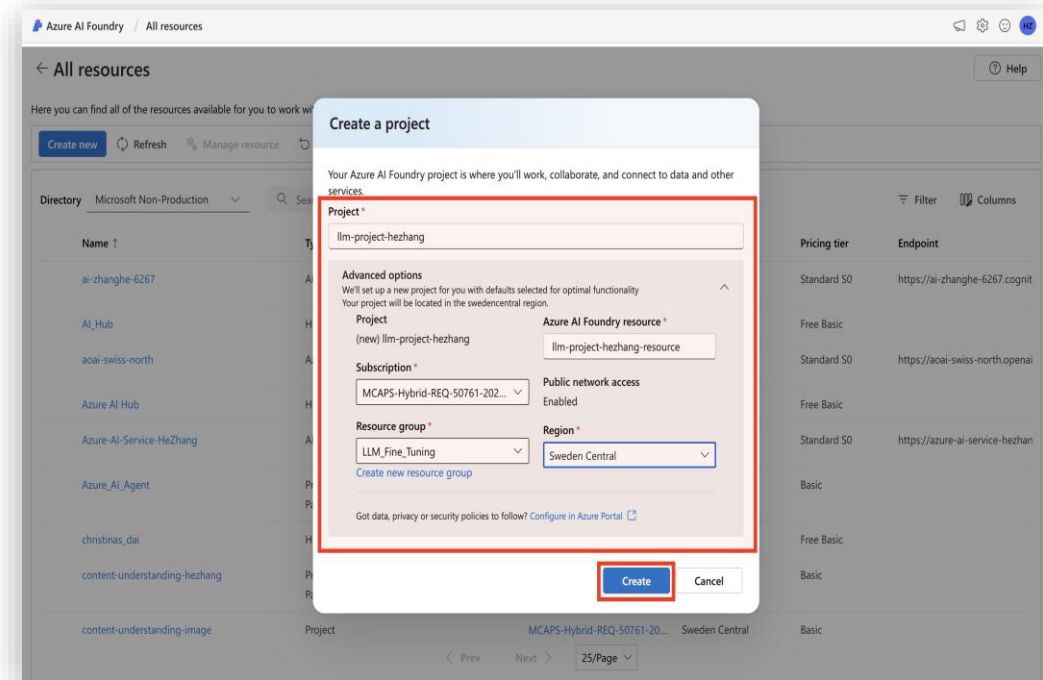
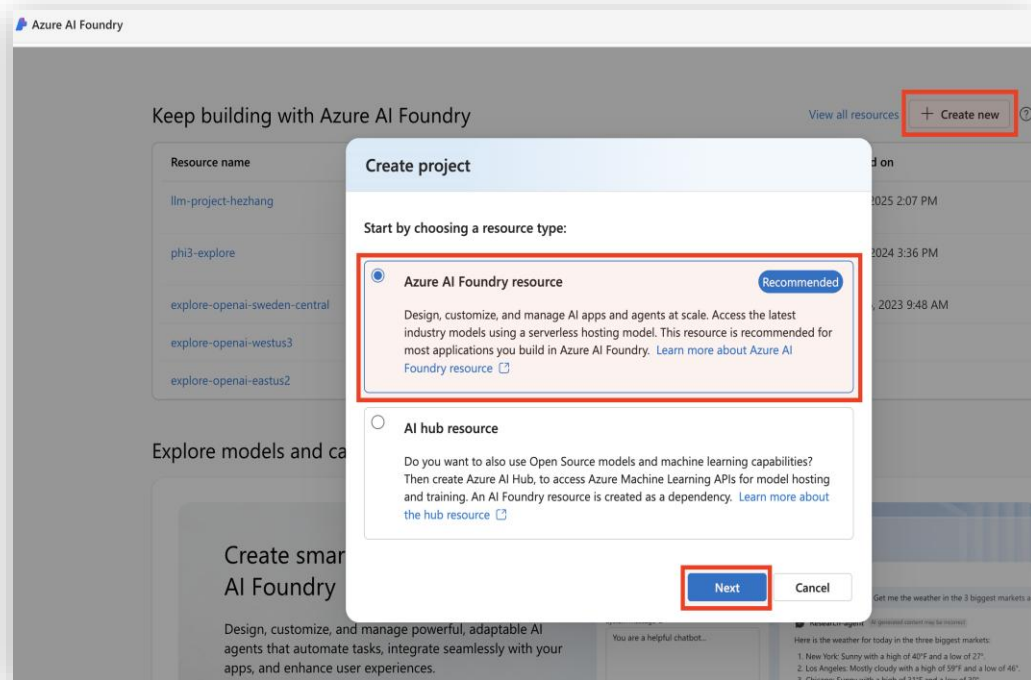
Optional Step: Continual Fine-Tuning



Fine-Tuning via Azure AI Foundry Dashboard

Fine-Tuning Open-Source Models

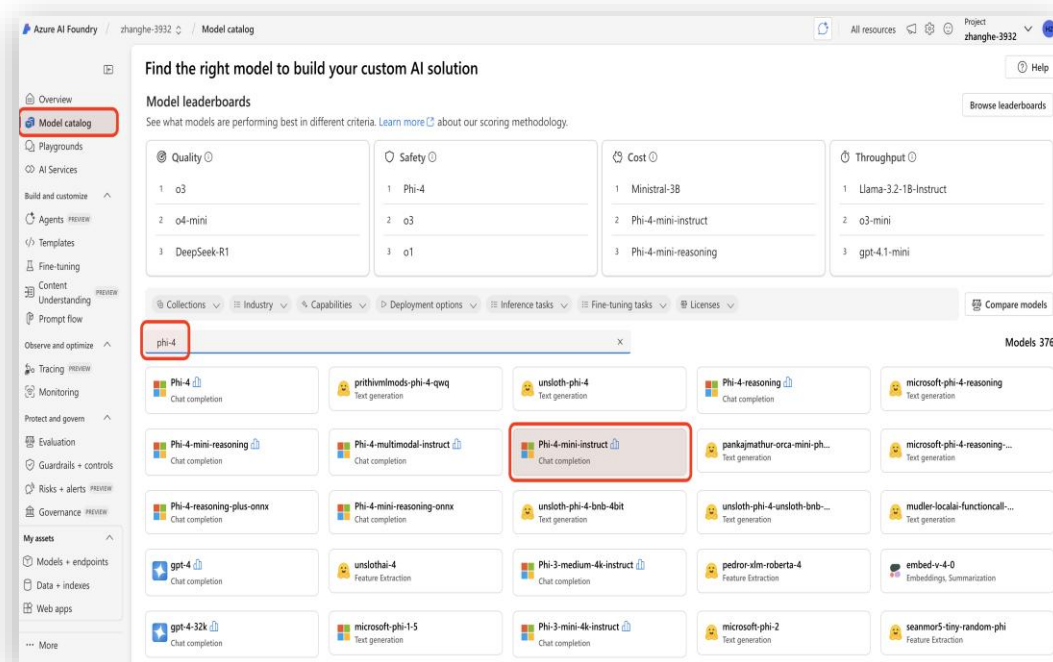
Step 1: Create a Project in Azure AI Foundry



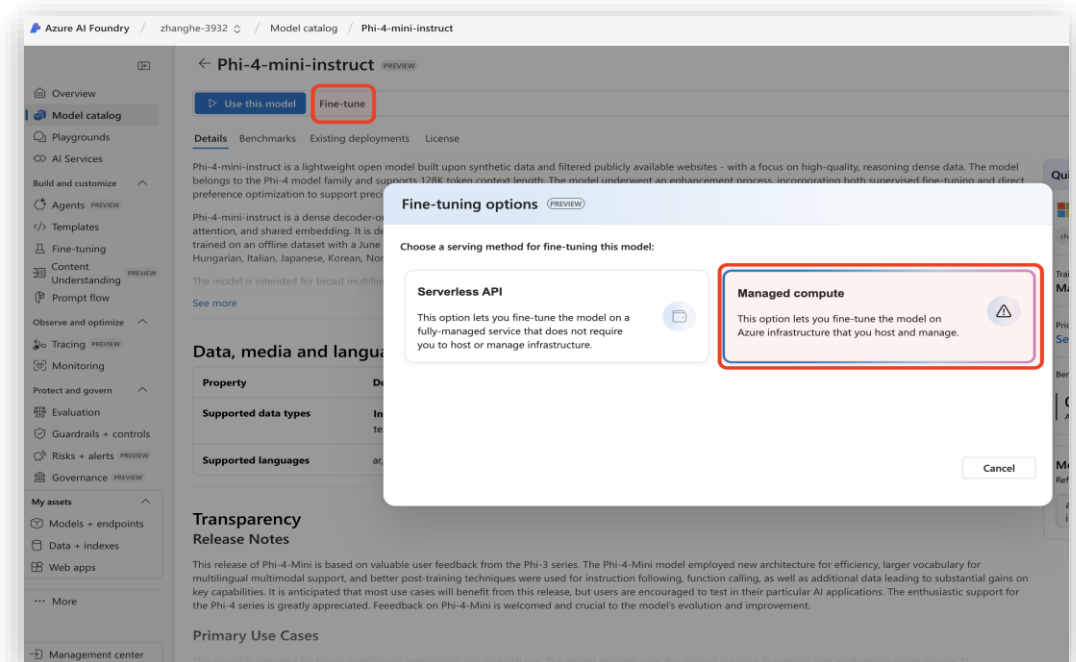
Fine-Tuning via Azure AI Foundry Dashboard

Fine-Tuning Open-Source Models

Step 2: Search *Model Catalog*



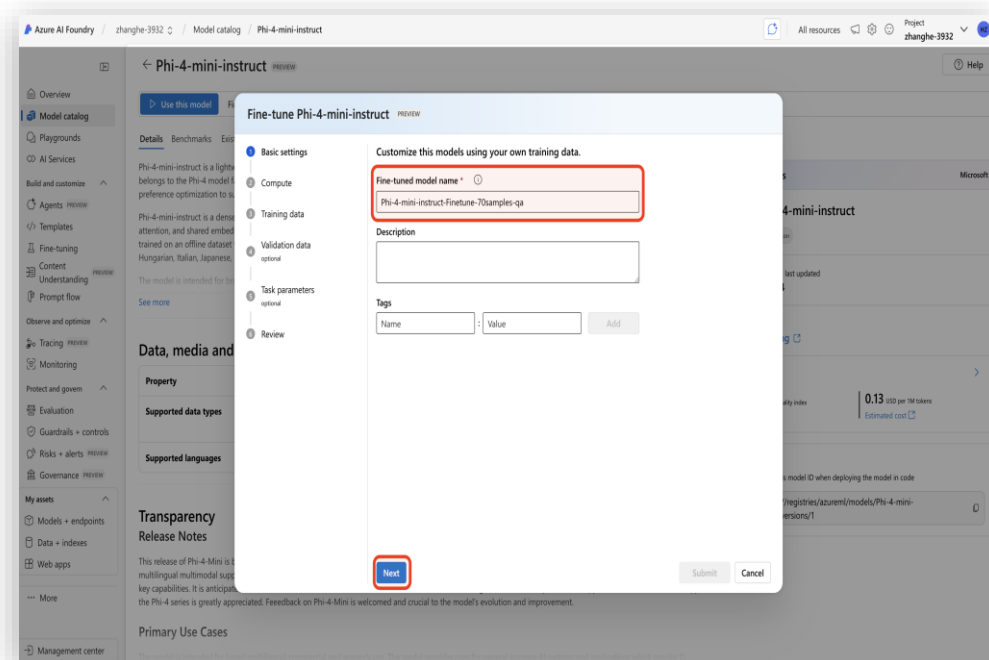
Step 3: Select *Managed compute*



Fine-Tuning via Azure AI Foundry Dashboard

Fine-Tuning Open-Source Models

Step 4: Give Basic Configs



Azure AI Foundry / zhanghe-3932 / Model catalog / Phi-4-mini-instruct

Phi-4-mini-instruct PREVIEW

Use this model

Fine-tune Phi-4-mini-instruct PREVIEW

Basic settings

Customize this models using your own training data.

Fine-tuned model name *

Phi-4-mini-instruct-Finetune-70samples-qa

Description

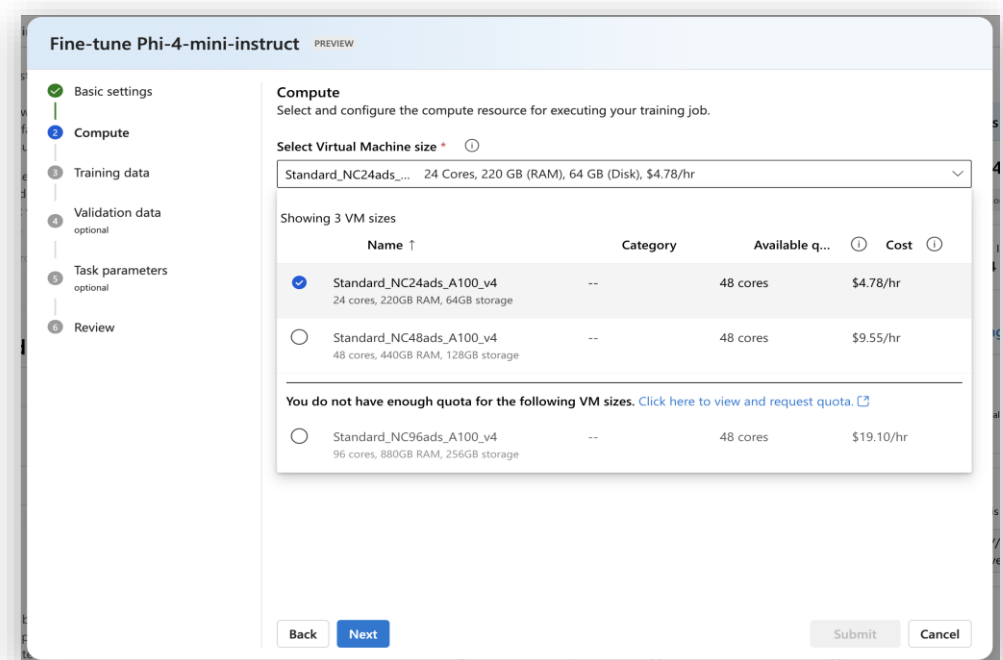
Tags

Name Value Add

Next

Submit Cancel

Step 5: Select Compute (GPU)



Fine-tune Phi-4-mini-instruct PREVIEW

Basic settings

Compute

Select and configure the compute resource for executing your training job.

Select Virtual Machine size *

Standard_NC24ads_v4 24 Cores, 220 GB (RAM), 64 GB (Disk), \$4.78/hr

Showing 3 VM sizes

Name ↑	Category	Available q...	Cost
☒ Standard_NC24ads_A100_v4 24 cores, 220GB RAM, 64GB storage	--	48 cores	\$4.78/hr
☐ Standard_NC48ads_A100_v4 48 cores, 440GB RAM, 128GB storage	--	48 cores	\$9.55/hr
☐ Standard_NC96ads_A100_v4 96 cores, 880GB RAM, 256GB storage	--	48 cores	\$19.10/hr

You do not have enough quota for the following VM sizes. [Click here to view and request quota.](#)

Back Next

Submit Cancel

Fine-Tuning via Azure AI Foundry Dashboard

Fine-Tuning Open-Source Models

Step 6: Upload Training Data

Fine-tune Phi-4-mini-instruct PREVIEW

Basic settings
Compute
3 Training data
4 Validation data optional
5 Task parameters optional
6 Review

Task type *
Chat completion

Select a dataset to personalize your model. Opt for a .jsonl file with prompt/completion pairs, ideally up to hundreds. [Learn more about preparing your own data](#)

Training data * ⓘ
Upload files

Choose a file to upload from your local drive. File must be under 200 MB.

Upload file

☐ Overwrite if already exists

Upload list

training_set_70samples.jsonl	✓ 18.5 KB/18.5 KB	...
------------------------------	-------------------	-----

messages

```
[{"role": "system", "content": "Clippy is a factual chatbot that is also sarcastic."}, {"role": "user", "content": "Who discover..."}, {"role": "system", "content": "Clippy is a factual chatbot that is also sarcastic."}, {"role": "user", "content": "What is the b..."}, {"role": "system", "content": "Clippy is a factual chatbot that is also sarcastic."}, {"role": "user", "content": "What is the la..."}
```

Back Next Submit Cancel

Step 7: Upload Validation Data (Optional)

Fine-tune Phi-4-mini-instruct PREVIEW

Basic settings
Compute
3 Training data
4 Validation data optional
5 Task parameters optional
6 Review

Validation data ⓘ Need help?

Select a dataset to personalize your model. Opt for a .jsonl file with prompt/completion pairs, ideally up to hundreds. [Learn more about preparing your own data](#)

Validation data * ⓘ
Upload files

Choose a file to upload from your local drive. File must be under 200 MB.

Upload file

☐ Overwrite if already exists

Upload list

validation_set_70samples.jsonl	✓ 18.06 KB/18.06 KB	...
--------------------------------	---------------------	-----

messages

```
[{"role": "system", "content": "Clippy is a factual chatbot that is also sarcastic."}, {"role": "user", "content": "What's the ca..."}, {"role": "system", "content": "Clippy is a factual chatbot that is also sarcastic."}, {"role": "user", "content": "Who wrote T..."}, {"role": "system", "content": "Clippy is a factual chatbot that is also sarcastic."}, {"role": "user", "content": "What's the de..."}
```

Back Next Submit Cancel

Fine-Tuning via Azure AI Foundry Dashboard

Fine-Tuning Open-Source Models

Step 8: Setup Parameters (Optional)

The screenshot shows the 'Task parameters' step in the 'Fine-tune Phi-4-mini-instruct' workflow. The left sidebar indicates the current step is 'Task parameters' (optional), with previous steps 'Basic settings', 'Compute', 'Training data', and 'Validation data' (optional) completed. The main area is titled 'Task parameters' and contains two sections: 'Preprocessing parameters' and 'Lora parameters'. In the 'Preprocessing parameters' section, 'Batch size' is set to 1000, 'Pad to max length' is toggled on, and 'Max sequence length' is set to -1. In the 'Lora parameters' section, 'Apply lora' is toggled on, 'Merge lora weights' is toggled off, 'Lora alpha' is set to 128, 'Lora r' is set to 8, and 'Lora dropout' is set to its default. At the bottom of the form are 'Back' and 'Next' buttons. A red rectangle highlights the entire 'Task parameters' section.

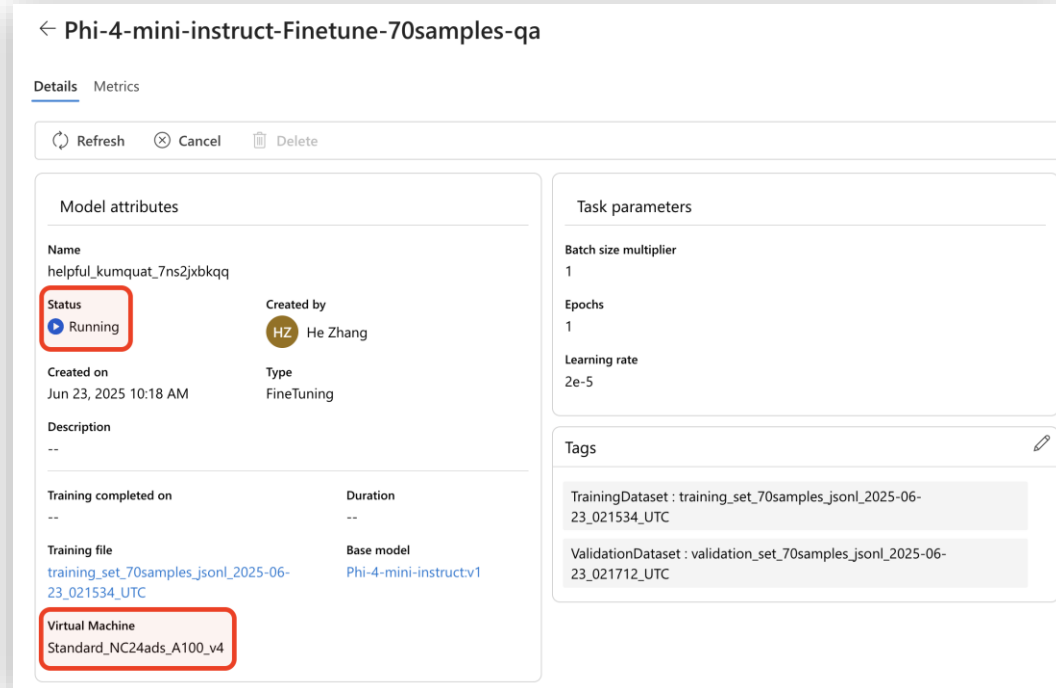
Step 9: Start Fine-Tuning Job

The screenshot shows the 'Review' step in the 'Fine-tune Phi-4-mini-instruct' workflow. The left sidebar indicates the current step is 'Review', with previous steps 'Basic settings', 'Compute', 'Training data', 'Validation data' (optional), and 'Task parameters' (optional) completed. The main area is titled 'Review' and contains a summary of the job configuration. It includes sections for 'Basic settings' (Fine-tuned model name: Phi-4-mini-instruct-Finetune-70samples-qa), 'Compute' (Compute Selected: Standard_NC24ads_A100_v4), and 'Task settings' (Task type: ChatCompletion, Training data: training_set_70samples_jsonl_2025-06-23_021534_UTC User data). At the bottom right, there are 'Back', 'Submit', and 'Cancel' buttons. A red rectangle highlights the 'Submit' button.

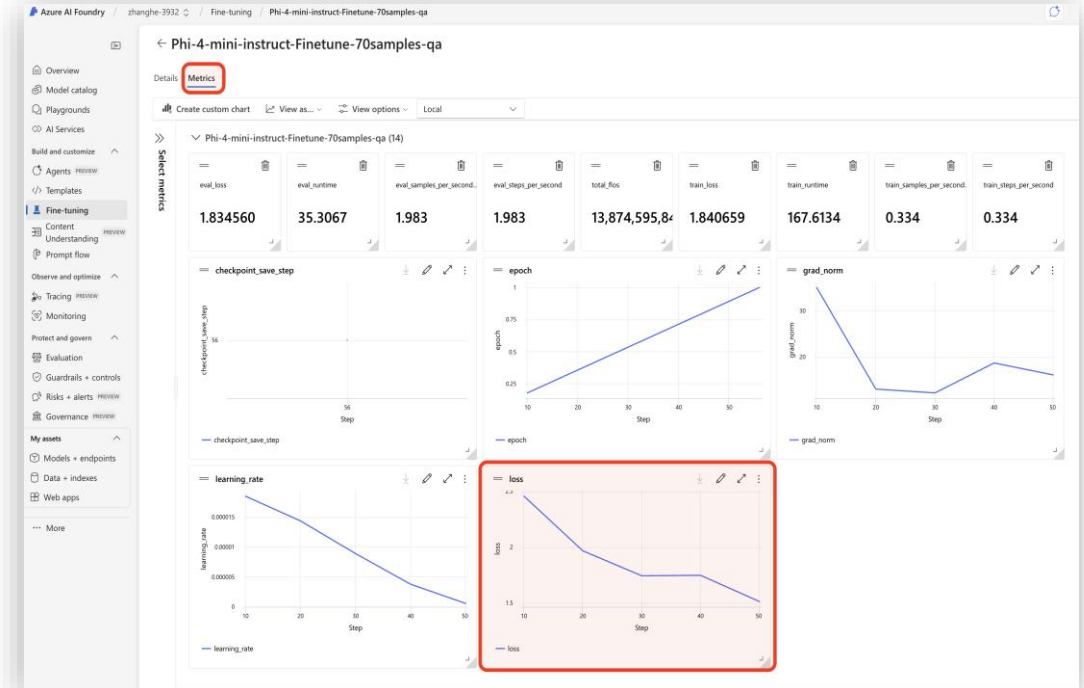
Fine-Tuning via Azure AI Foundry Dashboard

Fine-Tuning Open-Source Models

Step 10: Check Fine-Tuning Status



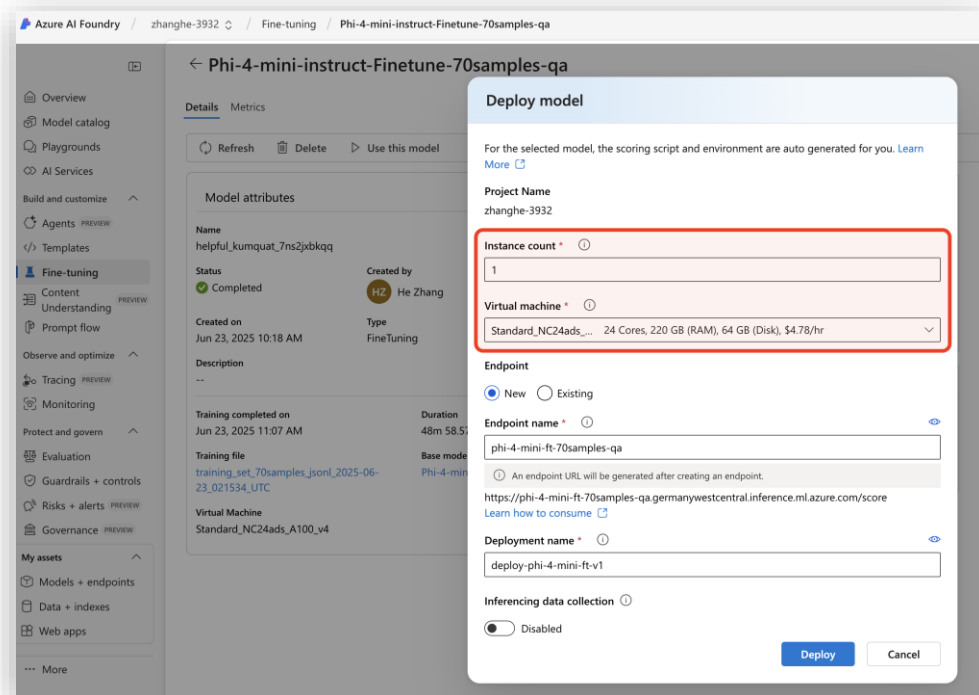
Step 11: Check Performance Metrics



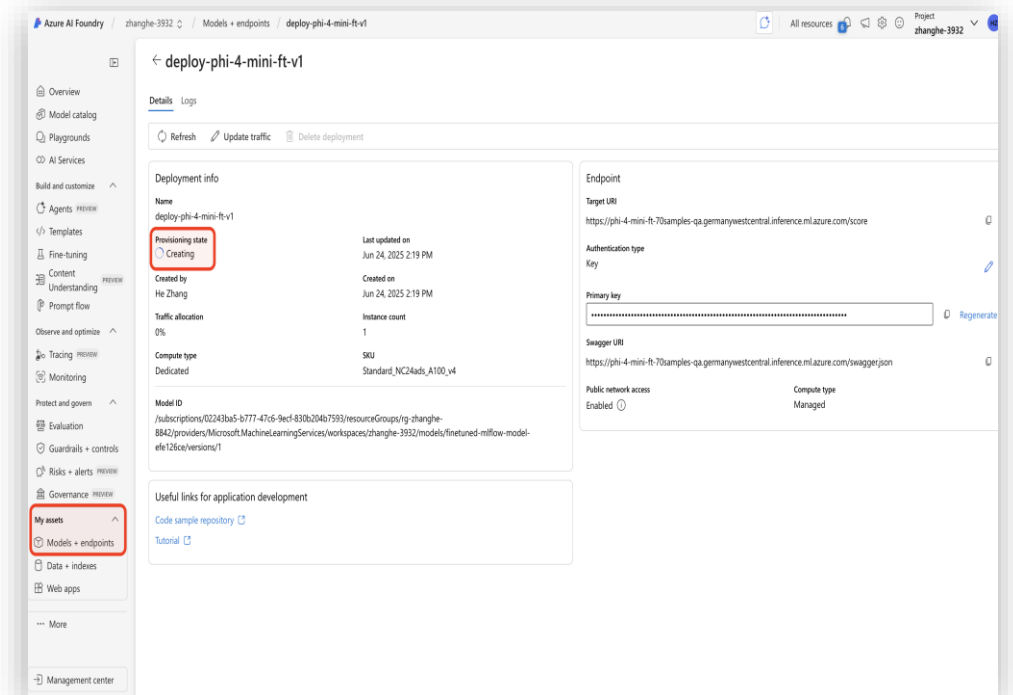
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Step 12: Deploy Fine-Tuned Model



Step 13: Check Deployment Status



Fine-Tuning via Azure AI Foundry Dashboard

Fine-Tuning Open-Source Models

Step 14: Test The Deployed Model via UI or API

The image displays three overlapping screenshots of the Azure AI Foundry dashboard, illustrating the process of testing a deployed model. The screenshots are arranged in a way that shows different parts of the interface, with red boxes highlighting specific areas of interest.

Top Screenshot: Shows the 'Details' tab for the model 'finetuned-mllow-model-efe126-1'. The 'Provisioning state' is highlighted with a red box and labeled 'Succeeded'. The 'Compute type' is also highlighted with a red box and labeled 'Standard_NC4ds_A100_v4'. The 'Endpoint' section shows the 'Target URI' as 'https://zhangle-3932-gpr1t.germanywestcentral.inference.ml.azure.com/score'.

Bottom Left Screenshot: Shows the 'Test' tab for the same model. The 'Sample inference' section is highlighted with a red box, showing the input data:

```
{  "input_data": {    "input_string": [      {        "role": "user",        "content": "I am going to China, what should I see?"      }    ],    "parameters": {      "max_new_tokens": 500    }  }  }
```

. The 'Test' button is also highlighted with a red box. Below this, the 'JSON Output' section is highlighted with a red box, showing a detailed response about visiting China.

Bottom Right Screenshot: Shows the 'Test' tab for the same model. The 'Basic consumption info' section is highlighted with a red box, showing the 'REST endpoint' as 'https://zhangle-3932-gpr1t.germanywestcentral.inference.ml.azure.com/score'. The 'Consumption option' section is also highlighted with a red box, showing the 'Consumption types' as 'JavaScript', 'Python', and 'JSON'. The 'Python' option is selected, and the corresponding code snippet is shown:

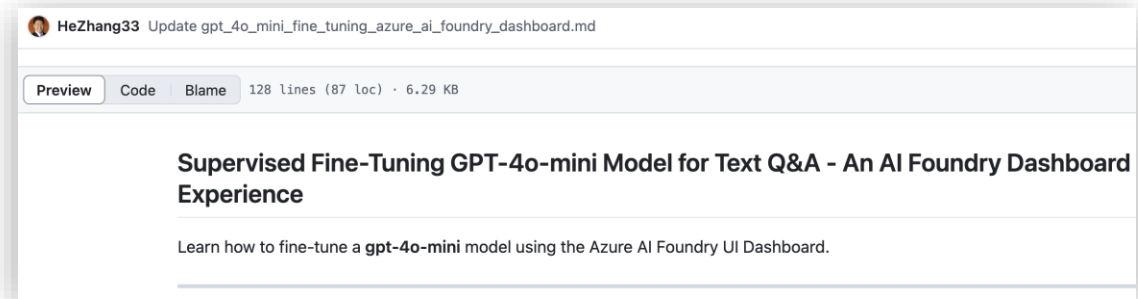
```
1 import urllib.request
2 import json
3
4 # Request data goes here
5 # The example below assumes JSON formatting which may be updated
6 # depending on the format your endpoint expects.
7 # More information can be found here:
8 # https://docs.microsoft.com/azure/machine-learning/how-to-deploy-advanced-entry-script
9 data = {}
10
11 body = str.encode(json.dumps(data))
12
13 url = 'https://zhangle-3932-gpr1t.germanywestcentral.inference.ml.azure.com/score'
14 req = urllib.request.Request(url, body, headers)
15 # Replace this with the primary/secondary key, mlToken, or Microsoft Entra ID token for the endpoint
16 # ml_key = "
17 raise Exception("A key should be provided to invoke the endpoint")
18
19
20 headers = {'Content-Type': 'application/json', 'Accept': 'application/json', 'Authorization': ('Bearer ', ml_key)}
21
22 req = urllib.request.Request(url, body, headers)
23
24 try:
25     response = urllib.request.urlopen(req)
26     result = response.read()
27     print(result)
28 }
```

Fine-Tuning via Azure AI Foundry Dashboard

Github Demos

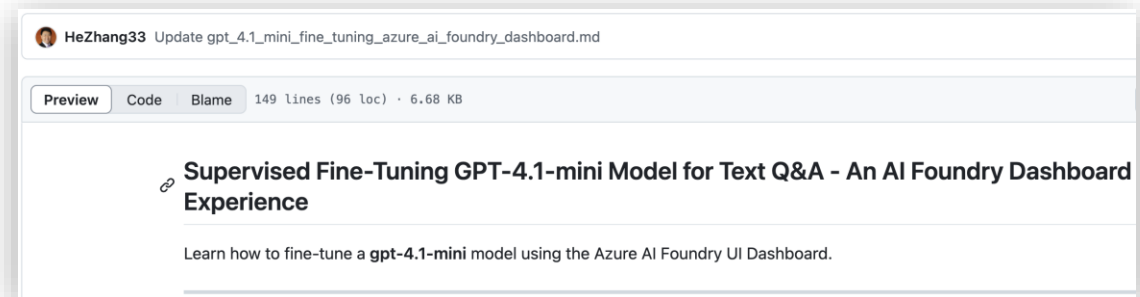
Fine-Tuning GPT-4o-mini:

https://github.com/microsoft/LLM-Fine-Tuning-Azure/blob/main/labs/fine_tuning_dashboards/gpt_4o_mini_fine_tuning_azure_ai_foundry_dashboard.md



Fine-Tuning GPT-4.1-mini:

https://github.com/microsoft/LLM-Fine-Tuning-Azure/blob/main/labs/fine_tuning_dashboards/gpt_4.1_mini_fine_tuning_azure_ai_foundry_dashboard.md



LLM Fine Tuning

Lab Description

- This lab outlines the process of fine-tuning the GPT-4.1-mini model using the Azure AI Foundry Dashboard, including dataset preparation, resource setup, model training, and deployment. It emphasizes a step-by-step workflow from signing into Azure, creating necessary resources, configuring the fine-tuning job with training and validation datasets, to reviewing metrics and testing the deployed model through the playground interface.

Questions?